Sequestration of Imaging Studies in MIDRC: Controlling for Ingenuous and Disingenuous Use of Sequestered Data

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Conflict of Interest Statement

- Dylan Tang: Nothing to Disclose
- Heather Whitney: Nothing to disclose
- Maryellen L. Giger: M.L.G is a stockholder in R2 technology/Hologic and QView, receives royalties from multiple medical imaging companies via licenses through the Polsky Center for Entrepreneurship and Innovation at the University of Chicago, and was a cofounder in Quantitative Insights.
- Kyle J. Myers: Puente Solutions LLC, works as an Independent Consultant to innovators seeking to gain US market authorization for medical imaging devices.

Introduction

The Medical Imaging and Data Resource Center (MIDRC)

MIDRC is

- A curated image and data commons
- A means by which researchers can address topics no single image archive could yield independently
- A partnership of the AAPM, ACR and RSNA, supported by NIBIB and ARPA-H, and hosted at University of Chicago on the Gen3 platform

There are two scientific components of MIDRC:

1. *Open Discovery Data Commons*

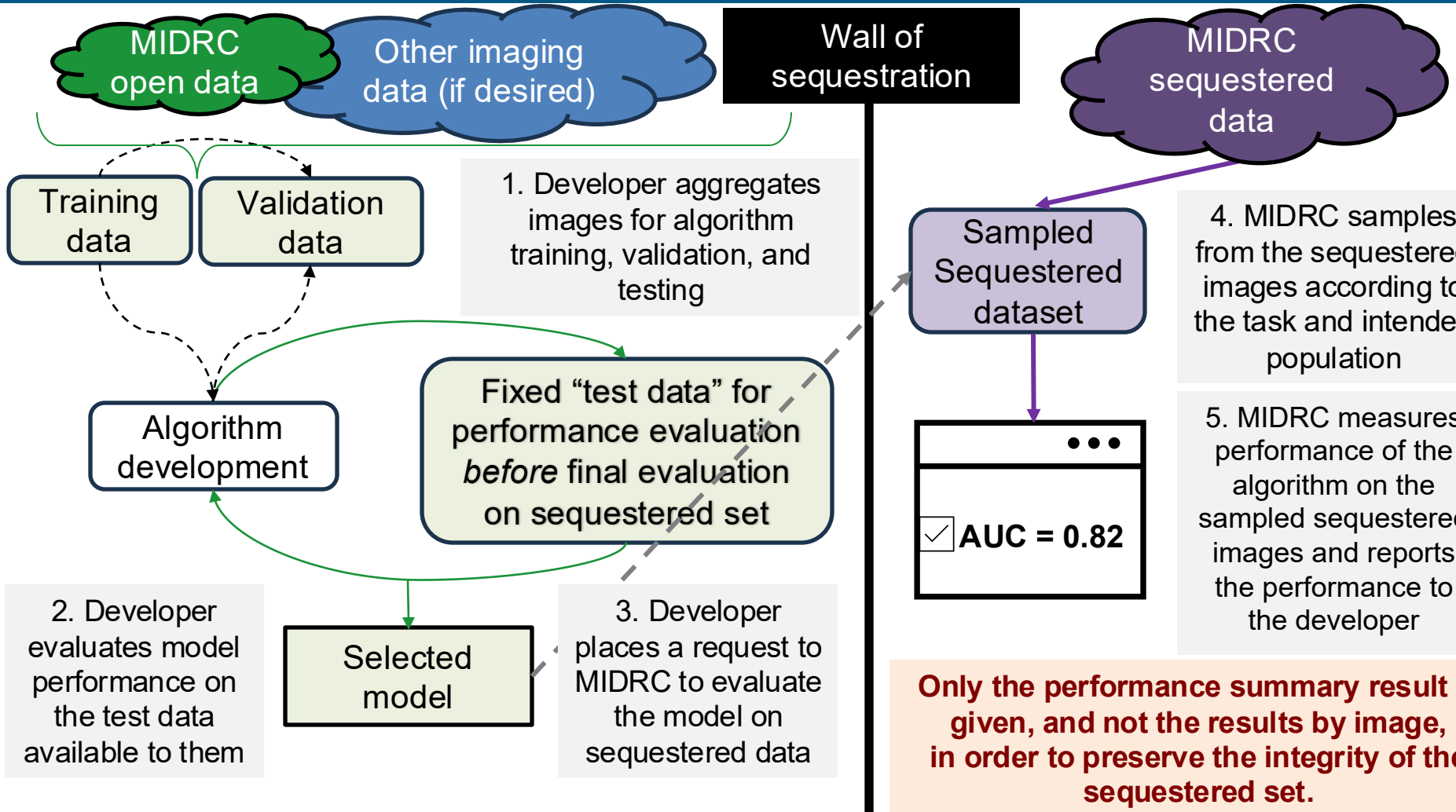
   Creation, testing, quality assurance, and data connectivity

2. *Machine Intelligence Computational Capabilities*

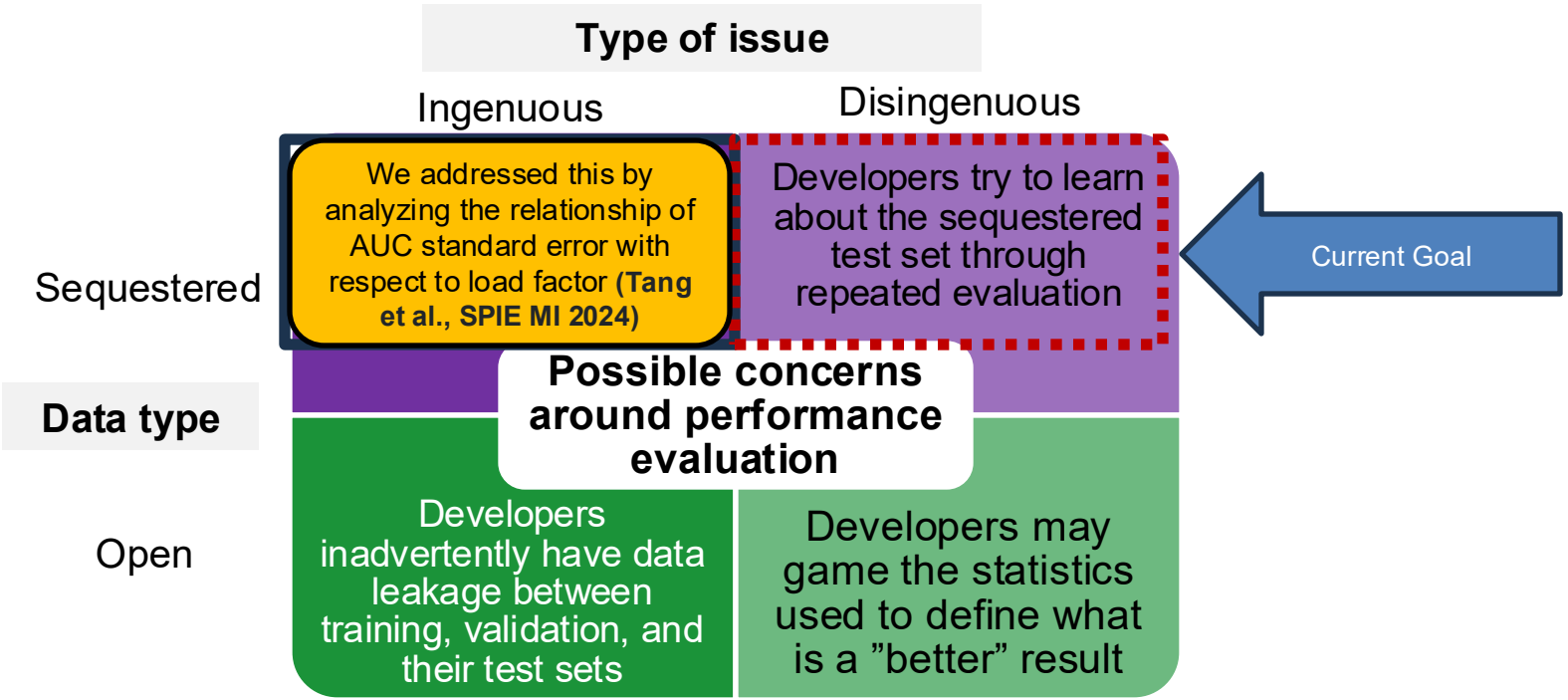
   Clinically relevant algorithms and software tools

Established August 2020
Supported by NIBIB

A collaboration of 24+ institutions & 100+ investigators from academia, community practices, FDA, and others


MIDRC open data
Other imaging data (if desired)
Wall of sequestration
MIDRC sequestered data
Training data
Validation data
1. Developer aggregates images for algorithm training, validation, and testing
Sampled Sequestered dataset
4. MIDRC samples from the sequestered images according to the task and intended population
Algorithm development
Fixed "test data" for performance evaluation *before* final evaluation on sequestered set
AUC = 0.82
5. MIDRC measures performance of the algorithm on the sampled sequestered images and reports the performance to the developer
2. Developer evaluates model performance on the test data available to them
Selected model
3. Developer places a request to MIDRC to evaluate the model on sequestered data
Only the performance summary result is given, and not the results by image, in order to preserve the integrity of the sequestered set.

Motivation



Dylan Tang, Heather M. Whitney, Kyle J. Myers, and Maryellen L. Giger "Sequestration of imaging studies in MIDRC: using load factor to minimize algorithm performance overestimation and image reuse", Proc. SPIE 12929, Medical Imaging 2024: Image Perception, Observer Performance, and Technology Assessment, 129290G (29 March 2024); https://doi.org/10.1117/12.3008264

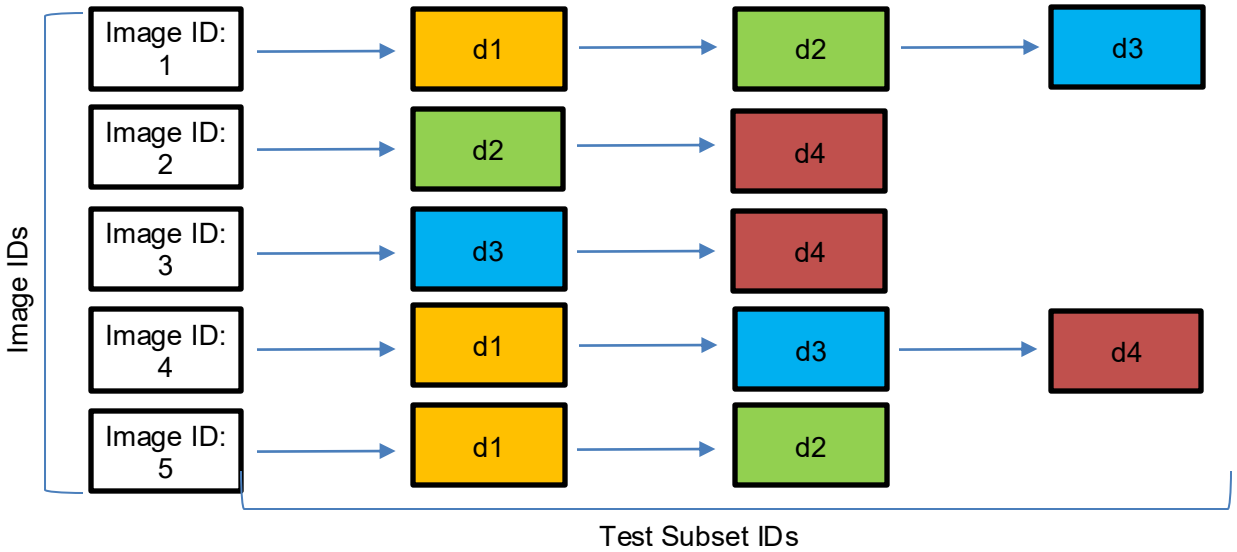
Motivation

- We investigate load factor, a metric that can be used to 'dial in' test subset distribution which avoids both ingenuous and disingenous use of the sequestered set
  1. Load factor controls for over-/under-estimation of AUC from evaluation of the sequestered test set
  2. Load factor controls for the reported number of test subset AUCs that differ from the AUC of the entire dataset

Background

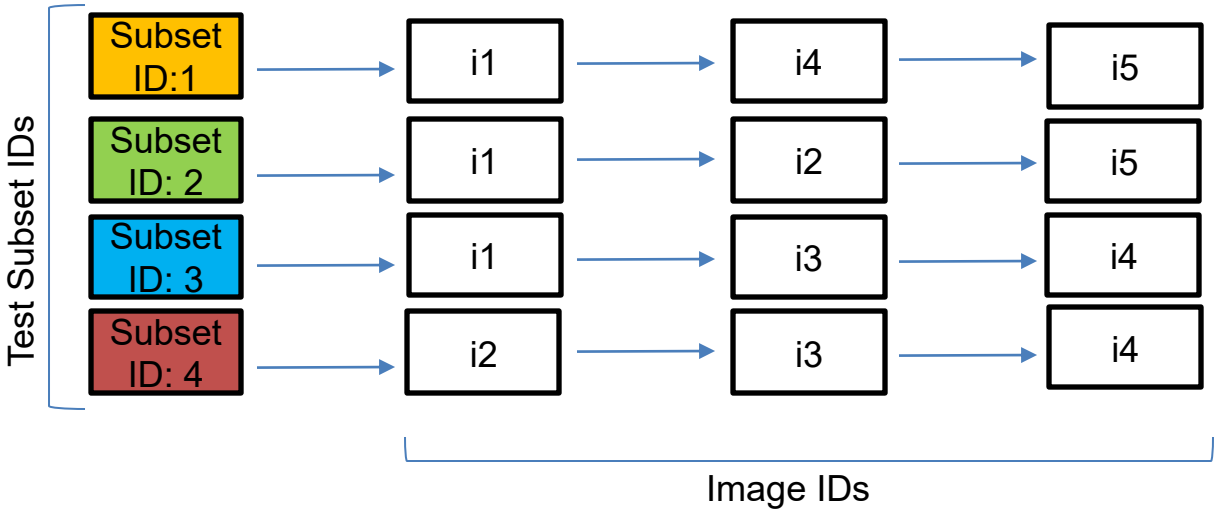
Hash Table Implementation

- Conceptual overview of a hash table for the example of mapping five images to four test subsets of size three



Hash Table Inversion

- Conceptual overview of hash table for the example of mapping five images to four test subsets of size three, but now inverted



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Load Factor

- The load factor is a ratio of test subsets to images and serves as a metric that describes the average number of images drawn per test subset.

- The load factor can serve as a metric that summarizes the state of the distribution process.

$$\text{load factor } (\xi) = \frac{\text{number of test subsets } (\rho) \cdot \text{test subset size } (N)}{\text{size of entire dataset } (\phi)}$$

High load factor: An image is likely used in **many** subsets

Low load factor: An image is likely used in **few** subsets

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$\text{Thresholdout}_{\text{AUC}}$

- $\text{Thresholdout}_{\text{AUC}}$ is an algorithm from *Gossman et al.* designed to mitigate potential dangers of reuse on an independent test set for an adaptive machine learning model.

- The algorithm involves perturbing the AUC of the test set by random noise if the absolute difference of the test set AUC is greater than the training set for each adaptively chosen ML model.

- For this study, we want to extend $\text{Thresholdout}_{\text{AUC}}$ to the hash table implementation mapping images to test subsets, and then evaluate the effects of the algorithm on a fixed ML model

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Objectives of the study

The purpose of our study was to develop a method to avoid 'disingenuous' use of the sequestered test set, i.e. by attempting to learn from the test set through repeat evaluations on test subsets:

- Adapt $\text{Thresholdout}_{AUC}$ to the hash table implementation mapping images to test subsets
- Devise analytical expressions between $\text{Thresholdout}_{AUC}$ parameters and load factor, a metric of fullness for the hash table
- Unite findings with previous work on 'ingenuous' use of the sequestered test set

Methods

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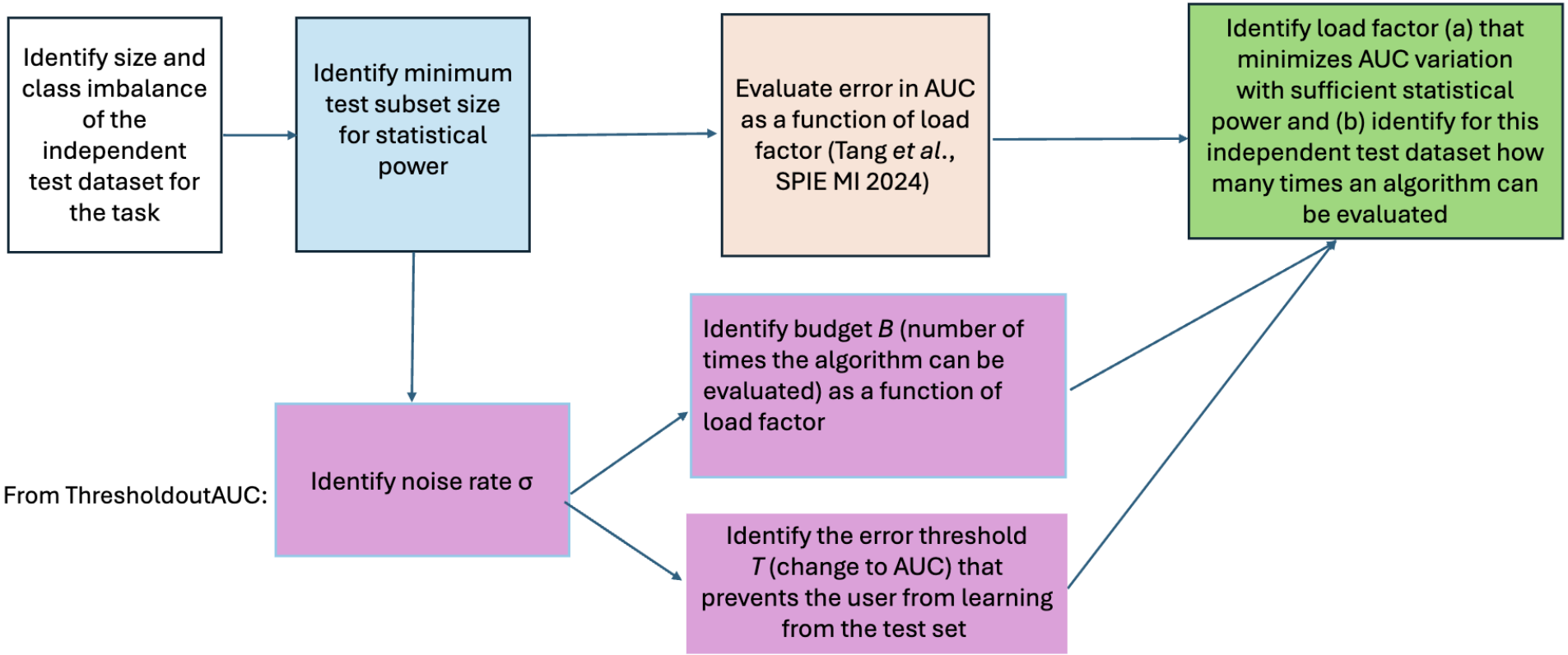
Roadmap of Methods



Methods: h-Thresholdout$_{AUC}$

procedure h-ThresholdOut$_{AUC}$ (S_{test}, *noiserate* σ, *budget* B, *threshold* T):

classifier $\phi \rightarrow [0, 1]$

for each test subset s_i in S_{test} and $B > 0$:

if $|AUC_{\phi_{s_i}} - AUC_{\phi}| > T$:

If the distance between AUC of the test subset and the AUC of the entire dataset exceeds T...

$\psi \sim Lap(\sigma)$

Randomly sample perturbation ψ from Laplace distribution and

OUTPUT($AUC_{\phi_{s_i}} + \psi$)

Output subset AUC + perturbation and

$B = B - 1$

Decrement budget by 1.

else

OUTPUT($AUC_{\phi_{s_i}}$)

Otherwise, return unperturbed test subset AUC

Methods: Analytical Relationship between Thresholdout$_{AUC}$ Budget and Load Factor

- Load factor (ξ) can be related to Thresholdout$_{AUC}$ budget with the following expression:

Load Factor

$$B = \frac{\sigma^5 \left(\frac{\xi \cdot \phi}{\rho}\right)^2}{512\left(\ln(8) + 2\sigma^2 \cdot \frac{\xi \cdot \phi}{\rho}\right)}$$

Thresholdout$_{AUC}$ Budget

Noise rate

where B is the Thresholdout$_{\mathrm{AUC}}$ budget, σ is the noise rate, ϕ is the number of images in the dataset, and ρ is the number of test subsets.

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Clinical Data Study

- A dataset of 1047 chest radiograph (CXR) images from COVID-19+ patients had been acquired between March and September of 2020.
- Each image was associated with the corresponding status of the patient as admitted to the intensive care unit (ICU) or not within 24 hours of imaging. 12% of patients had been admitted to the ICU.
- A previously-trained deep learning/artificial intelligence model using DenseNet121 architecture had been used to predict patients' need for intensive care within 24 hours of imaging.
- The area under the receiver operating characteristic curve (AUC) in the task of predicting patient ICU admission within 24 hours was AUC = 0.81.

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Methods: Evaluation

From the clinical data study, we hope to establish the following:

(1) Statistical power as a function of test subset size and load factor

- Goal: to determine a minimum bound on test subset size that has sufficient statistical power, given a particular load factor

(2) Distribution of $\text{Thresholdout}_{AUC}$ perturbations at different load factors

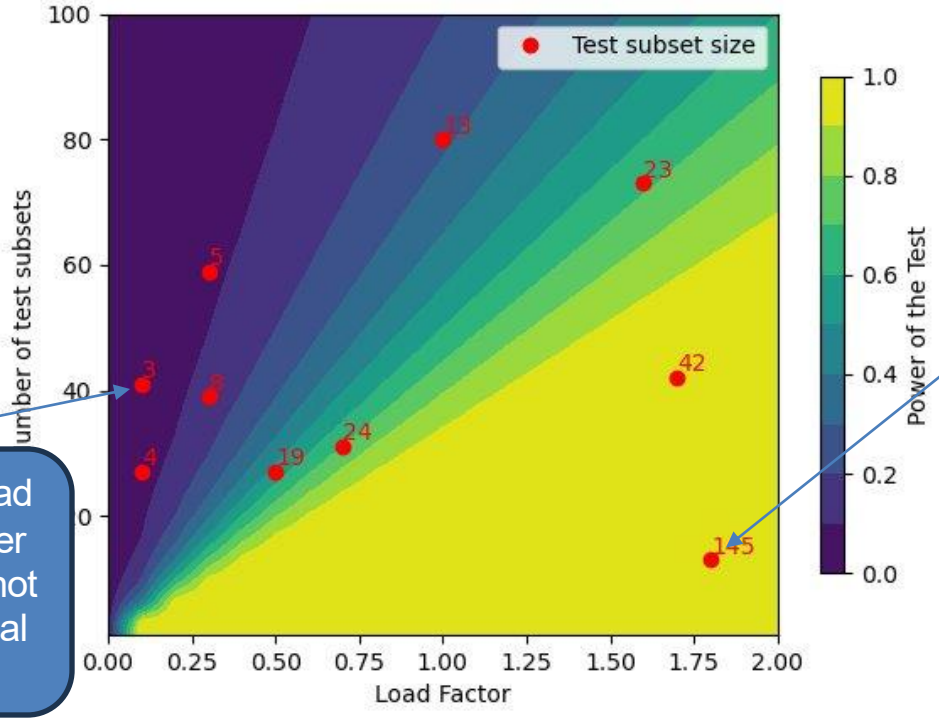
- Goal: to evaluate how the frequency of perturbed test subset AUCs changes with load factor and test subset size

(3) Load factor versus $\text{Thresholdout}_{AUC}$ budget with respect to noise rate

- Goal: to establish load factor as a metric that can control for $\text{Thresholdout}_{AUC}$ budget and AUC variation

Results

Results: (1) Statistical Power and Minimum Test Subset Size



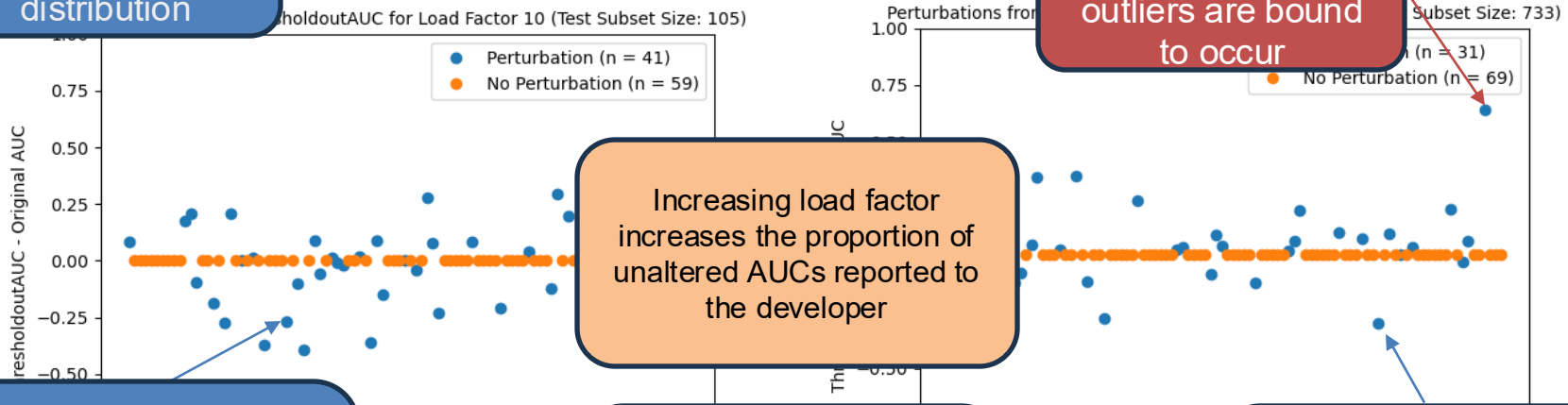
Operating at a low load factor and low number of test subsets does not ensure good statistical power

Operating at a high load factor and a high number of test subsets typically ensures good statistical power

Results: (2) Distribution of $\text{Thresholdout}_{AUC}$ Perturbations

Noise rate σ controls for the variability of the distribution

Due to random sampling, some outliers are bound to occur



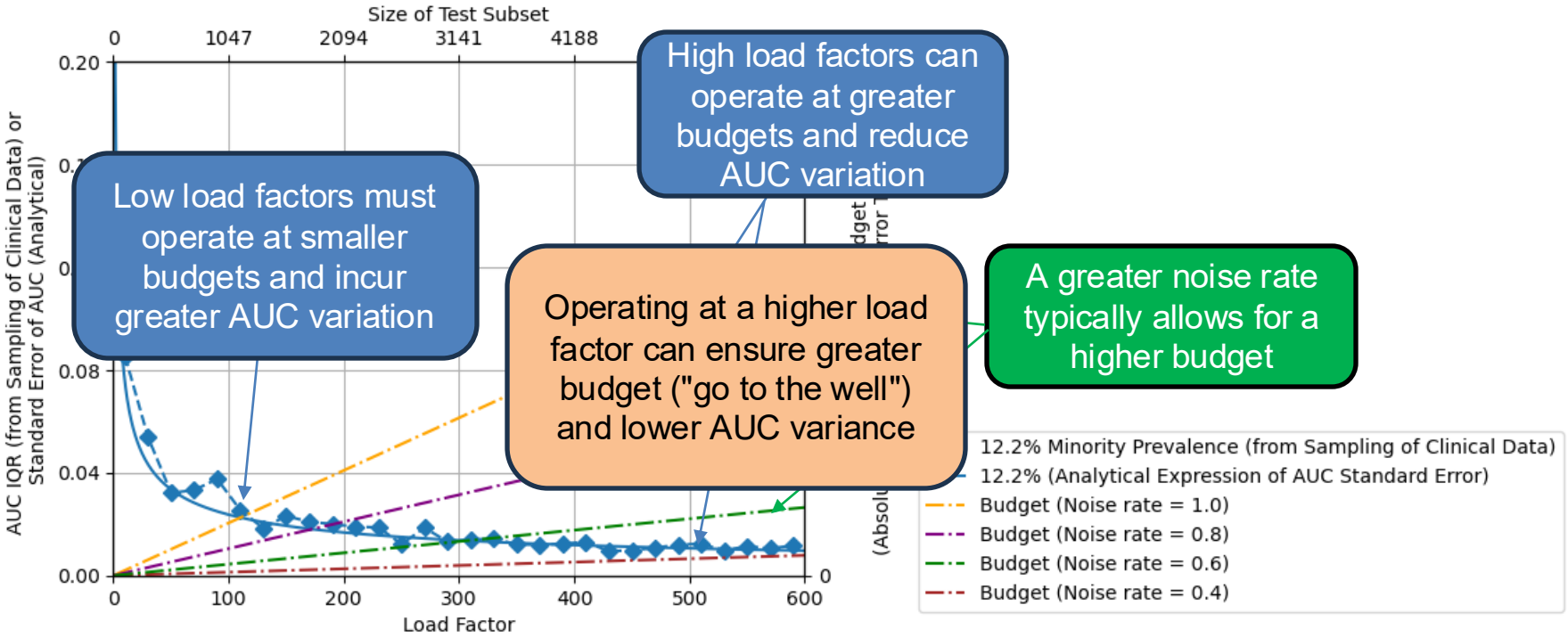
Increasing load factor increases the proportion of unaltered AUCs reported to the developer

At a load factor of 10, around 60% of test subsets have an AUC that will be perturbed by $\text{Thresholdout}_{AUC}$

The distributions were generated with a noiserate σ = 0.124

At a load factor of 70, around 40% of test subsets have an AUC that will be perturbed by $\text{Thresholdout}_{AUC}$

Results: (3) Relationship between AUC Variation and Thresholdout$_{AUC}$ budget



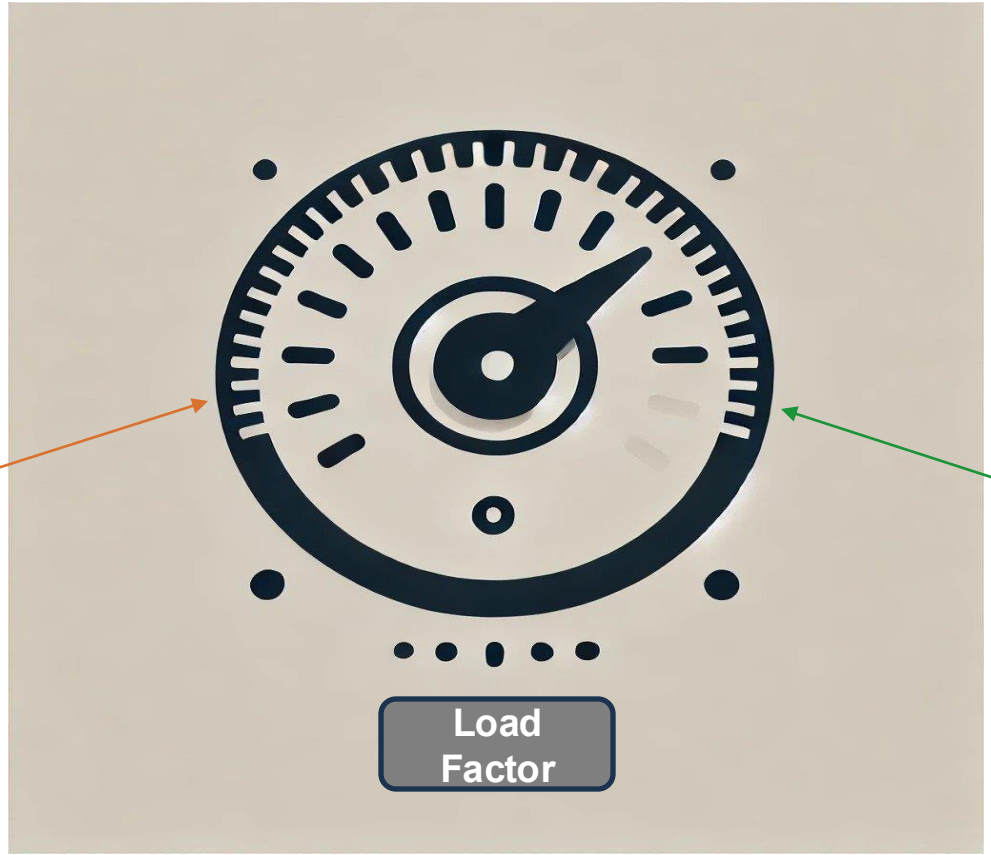
High load factors can operate at greater budgets and reduce AUC variation

Low load factors must operate at smaller budgets and incur greater AUC variation

Operating at a higher load factor can ensure greater budget ("go to the well") and lower AUC variance

A greater noise rate typically allows for a higher budget

A Heuristic for Load Factor



Low Load Factor:
High AUC Variation,
Low Budget

High Load Factor:
Low AUC Variation,
High Budget

Conclusion

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Conclusion

From our results, we conclude that the load factor metric, when examined with the $\text{Thresholdout}_{AUC}$ algorithm adapted to the hash table implementation is:

1. A regulating metric that can identify the frequency of $\text{Thresholdout}_{AUC}$ perturbations alongside ensuring statistical power
2. A metric that can simultaneously constrain $\text{Thresholdout}_{AUC}$ budget and AUC variation.

These results show that load factor is a potential metric that can control for both ingenuous and disingenuous use of a dataset.

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